



Fooling LIME and SHAP: Adversarial Attacks on Post hoc Explanation Methods

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Motivation

- ML has been applied for **critical** decision making
 - Healthcare
 - Criminal justice
 - Finance
- The decision makers must clearly **understand the model behavior** to
 - Diagnose the error and potential biases
 - Decide when and how much these ML models should be trusted

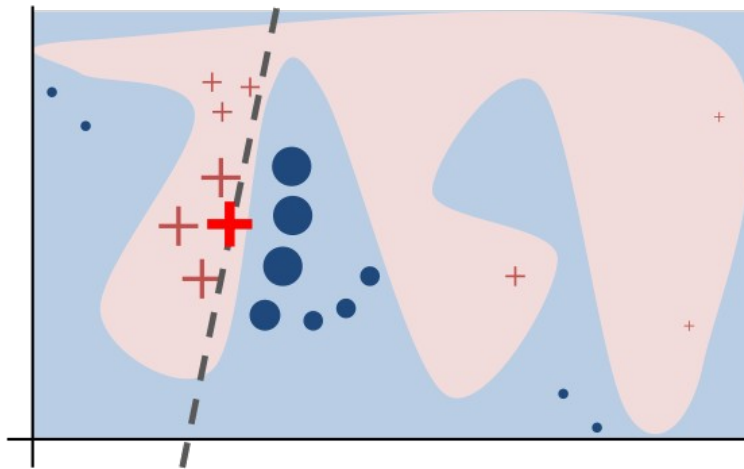
Motivation

- Trade-off between interpretability and accuracy
 - **Simple** models can be easily **interpreted** (e.g., linear regression)
 - **Complex** but also black-box model has much **better performance** (e.g., deep neural network)
- Can a ML method be both **interpretable** and **accurate**?
- **Post hoc** explanation can seemingly solve this problem:
 - First build **complex and accurate** ML models for good performance
 - Then use post hoc explanation for model **interpretation**
- The question is:
 - How robust and reliable is the *post hoc* explanation methods?

Contribution: A framework to ‘fool’ the post hoc explanation method

- A novel framework that can effectively **mask the discriminatory biases** of any black box classifier
 - Fooling the **perturbation based** post hoc explanation method
 - LIME and SHAP
- Allowing an adversarial entity to control and generate an arbitrary desired explanation
- Demonstration using real-world datasets with extremely biased classifier
- Existing post hoc explanation techniques are **NOT** sufficiently robust for ascertaining discriminatory behavior of classifiers in sensitive applications

Perturbation-based post hoc explanation method



Preliminaries & Background

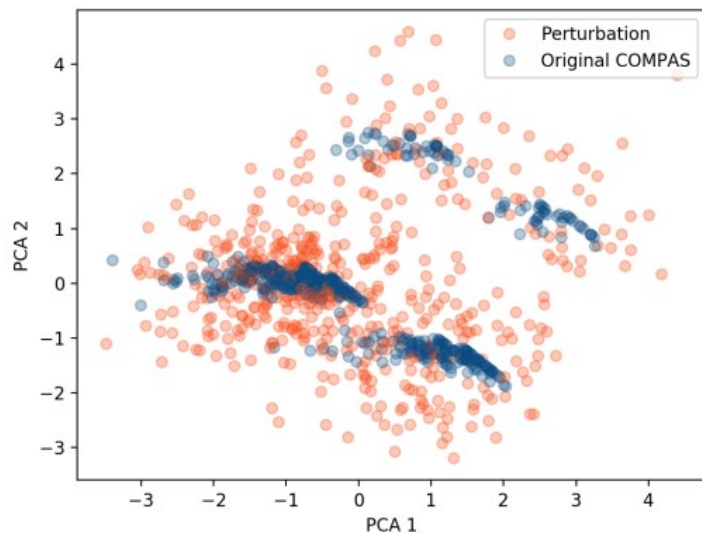
$$\arg \min_{g \in \mathcal{G}} L(f, g, \pi_x) + \Omega(g)$$

where the loss function L is defined as:

$$L(f, g, \pi_x) = \sum_{x' \in X'} [f(x') - g(x')]^2 \pi_x(x')$$

- f is the original classifier and x is the datapoint we want to explain
- g is the explanation we want to learn, $\Omega(g)$ is the “complexity” of g
- π is the proximity measure
- X' is a **synthetic dataset**, consisting of perturbations of x

Intuition





Approach: Set-up

Adversary would like to deploy a biased classifier f !

- Background: the biased model f uses sensitive attributes to make critical decisions
- Requirement: give access of black-box models to customers and regulators who use post-hoc explanations
- Goal: hide bias of the classifier f

Approach: Set-up

What do we need?

- Input: dataset sampled from real-world distribution
- Target Product: an adversarial classifier e
 - f is the biased model to be explained, while ψ is an unbiased model

$$e(x) = \begin{cases} f(x), & \text{if } x \in \mathcal{X}_{dist} \\ \psi(x), & \text{otherwise} \end{cases}$$

Approach: OOD Detection

Which of the inputs belong to the real-world distribution?

- Build another classifier for OOD detection
 - Assign label “False” (not OOD) to all instance in the dataset $\mathbf{X}$
 - Perturb all instances in $\mathbf{X}$ and assign them label “True” (OOD)
 - Exceptions: instances too close to observations from $\mathbf{X}$
 - Combine data and train OOD detection classifier

$$e(x) = \begin{cases} f(x), & \text{if } x \in \mathcal{X}_{dist} \\ \psi(x), & \text{otherwise} \end{cases}$$

Experiment: Set-up

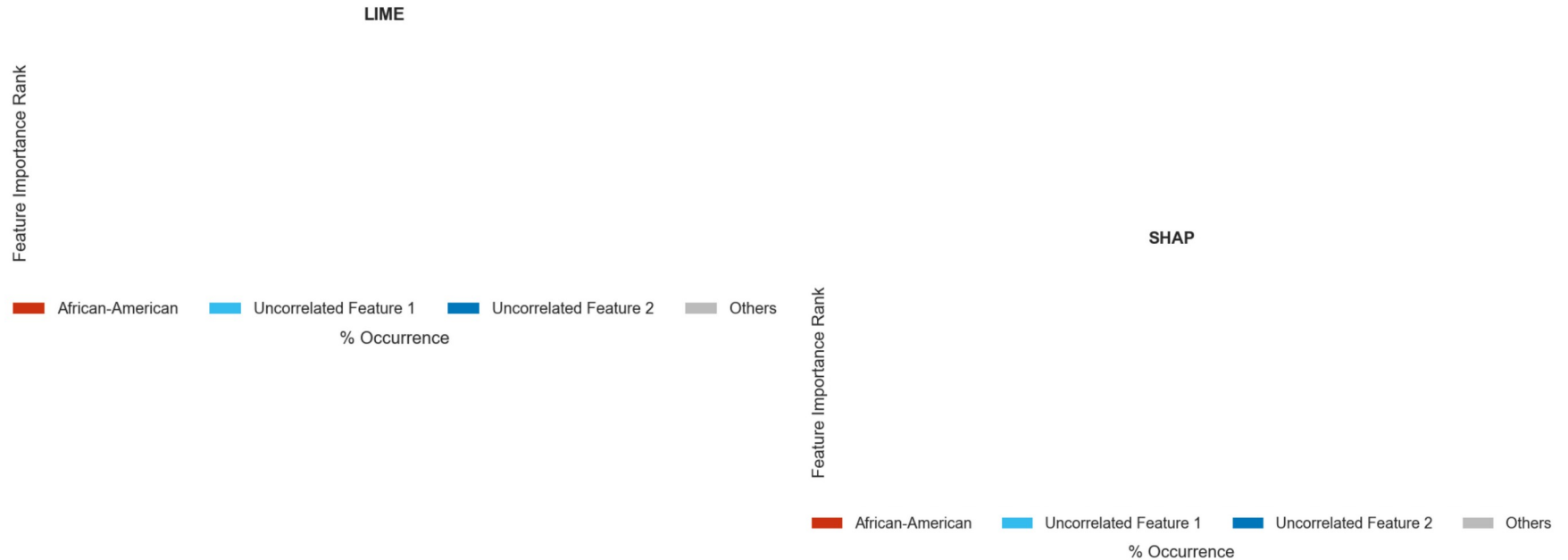
90% training & 10% test

Dataset	Size	Features	Positive Class	Sensitive Feature
COMPAS	6172	<i>criminal history, demographics, COMPAS risk score, jail and prison time</i>	High Risk (81.4%)	African-American (51.4%)
Communities & Crime	1994	<i>race, age, education, police demographics, marriage status, citizenship</i>	Violent Crime Rate (50%)	White Population (continuous)
German Credit	1000	<i>account information, credit history, loan purpose, employment, demographics</i>	Good Customer (70%)	Male (69%)

Biased classifier f makes predictions purely based on sensitive attributes (race, gender)

Unbiased classifier ψ uses only features uncorrelated with the sensitive attributes

Experiment: Results - COMPAS



Experiment: Results - Communities and Crime

LIME

Feature Importance Rank

Race % White Unrelated Feature 1 Unrelated Feature 2 Others
% Occurrence

SHAP

Feature Importance Rank

Race % White Unrelated Feature 1 Unrelated Feature 2 Others
% Occurrence

Experiment: Results - German credit

Feature Importance Rank

Gender Loan Rate % Income All Others
% Occurrence

Takeaway from experiments

1. Accuracy of the OOD classifier -> success of the adversarial attack
2. Adversarial classifiers to LIME are ineffective against SHAP explanations
 - a. Any sufficiently accurate OOD classifier is sufficient to fool LIME, while fooling SHAP requires more accurate classifiers
3. SHAP less successful when using two features <- local accuracy property
 - a. Distribute attributions among several features

$$e(x) = \begin{cases} f(x), & \text{if } x \in \mathcal{X}_{dist} \\ \psi(x), & \text{otherwise} \end{cases}$$



Conclusions

- Main contribution: A framework for converting any black-box classifier into a *scaffolded* classifier that fools perturbation-based post-hoc explanation techniques like LIME and SHAP
- Effectiveness of this framework demonstrated on sensitive real-world data (criminal justice and credit scoring)
- Perturbation-based post-hoc explanation techniques are not sufficient to test whether classifiers discriminate based on sensitive attributes



Related Works

- Issues with post-hoc explanations:
 - [Doshi-Velez and Kim] identify explainability of predictions as a potentially useful feature of interpretable models.
 - [Lipton] and [Rudin] argues post-hoc explanations can be misleading and are not trustworthy for sensitive applications.
 - [Ghorbani et al.] and [Mittelstadt et al.] identified further weaknesses of post-hoc explanations.
- Adversarial explanations
 - [Dombrowski et al.] and [Heo et al.] show how to change saliency maps in arbitrary ways by imperceptibly changing inputs.



Q&A

- Are the experimental results sufficient to justify the conclusions?
 - In particular, how can we explain the discrepancy in results for LIME vs. SHAP?
- What about fooling other classes of post-hoc explanation methods?
 - Past work: gradient-based methods
- Alternatively, can one design post-hoc explanations that are *adversarially robust*?



Explanations can be manipulated and geometry is to blame

Authors: Ann-Kathrin Dombrowski, Maximilian Alber, Christopher J. Anders, Marcel Ackermann, Klaus-Robert Müller, Pan Kessel

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Introduction

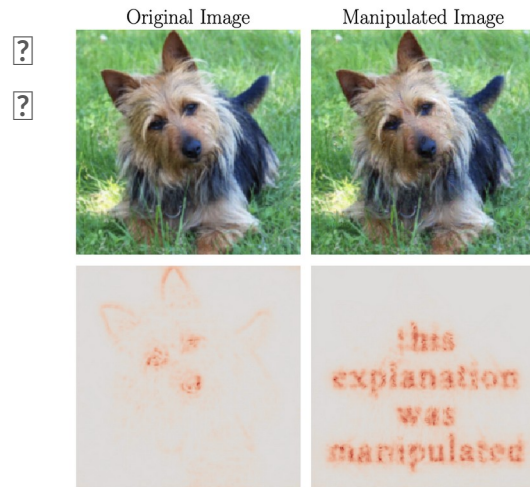
Motivation

Understand and verify aspects of ML models
Aid decision making in high-stakes scenarios } → **Reliable** explanations of models!

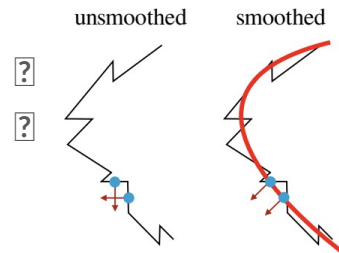
Can we always trust model
explanations?

Summary / Contribution

- Manipulate explanations!
- Provide a theoretical understanding of such nonrobustness and derive a bound
- Introduce smoothing to increase explanation robustness!



$$\|h(p) - h(p_0)\| \leq |\lambda_{max}| d_g(p, p_0) \leq \beta C d_g(p, p_0),$$



Background + Related Work

- Interpretation of Neural Networks is Fragile [[Ghorbani et al.](#)]
 - Complex decision boundary
- The (un)reliability of saliency methods [[Kindermans et al.](#)]
 - Input invariance
- Sanity checks for saliency maps [[Adebayo et al.](#)]
 - Randomization test (Wednesday)
- Fairwashing Explanations with Off-Manifold Detergent [[Anders et al.](#)]
 - Low-dimensional data manifold v.s. High-dimensional embedding space

Methodology

Notation

- Neural network $g : \mathbb{R}^d \rightarrow \mathbb{R}^K$ with relu non-linearities
- Classifies input image x into K categories, predicted class $k = \arg \max_i g(x)_i$
- Explanation map: $h : \mathbb{R}^d \rightarrow \mathbb{R}^d$
- Target map: $h^t \in \mathbb{R}^d$
- Manipulated image: $x_{adv} = x + \delta x$



Properties of Manipulated Image

1. The output of the network stays approximately constant, i.e. $g(x_{adv}) \approx g(x)$.
2. The explanation is close to the target map, i.e. $h(x_{adv}) \approx h^t$.
3. The norm of the perturbation δx added to the input image is small, i.e. $\|\delta x\| = \|x_{adv} - x\| \ll 1$ and therefore not perceptible.

Explanation Methods

Gradient-based

- Vanilla gradients: $h(x) = \frac{\partial g}{\partial x}(x)$
 - Quantifies how infinitesimal perturbations in each pixel change the prediction
- Gradient $\times$ Input: $h(x) = x \odot \frac{\partial g}{\partial x}(x)$
 - For linear models, this measure gives the exact contribution of each pixel to the prediction
- Integrated Gradients: $h(x) = (x - \bar{x}) \odot \int_0^1 \frac{\partial g(\bar{x} + t(x - \bar{x}))}{\partial x} dt$

Propagation-based

- Guided Backpropagation
- Layer-wise Relevance Propagation
- Pattern Attribution
 - Standard backpropagation upon element-wise multiplication of the weights with learned patterns

Manipulation Method

- Obtain manipulated images by optimizing the loss function

$$\mathcal{L} = \|h(x_{adv}) - h^t\|^2 + \gamma \|g(x_{adv}) - g(x)\|^2$$

manipulated
explanation
map

target
map

weighting
hyperparameter

network output
(manipulated
input)

network output
(original input)

with respect to x_{adv} using gradient descent

Manipulation Method

- The gradient with respect to the input $\nabla h(x)$ of the explanation often depends on the vanishing second derivative of the relu non-linearities. This causes problems during optimization of the loss function:

$$\partial_{x_{adv}} \|h(x_{adv}) - h^t\|^2 \propto \frac{\partial h}{\partial x_{adv}} = \frac{\partial^2 g}{\partial x_{adv}^2} \propto \text{relu}'' = 0$$

- **Solution:** replace relu with softplus

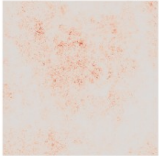
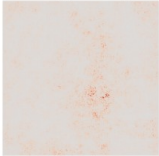
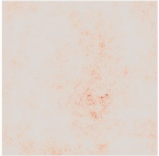
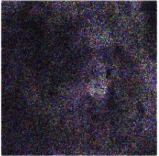
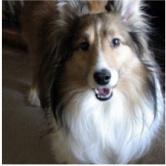
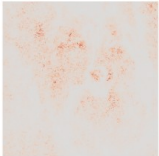
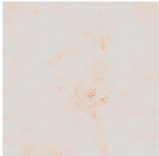
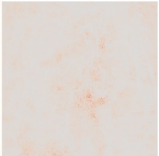
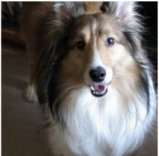
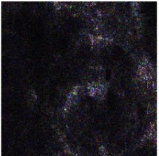
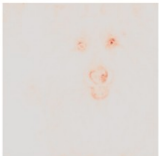
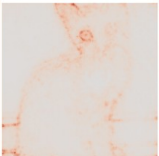
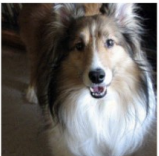
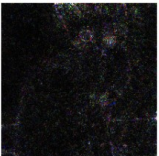
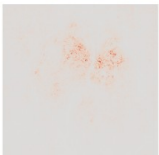
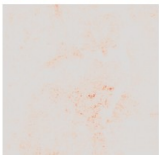
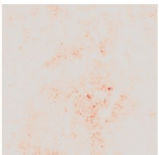
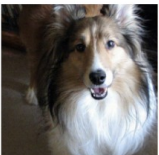
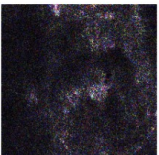
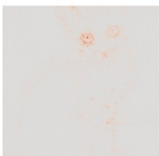
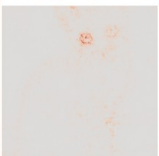
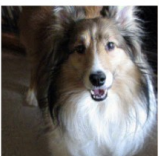
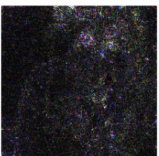
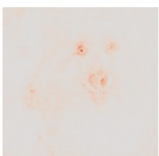
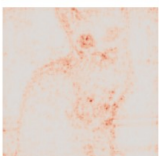
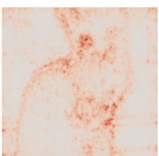
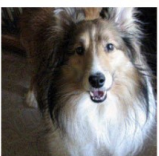
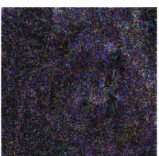
$$\text{softplus}_{\beta}(x) = \frac{1}{\beta} \log(1 + e^{\beta x})$$

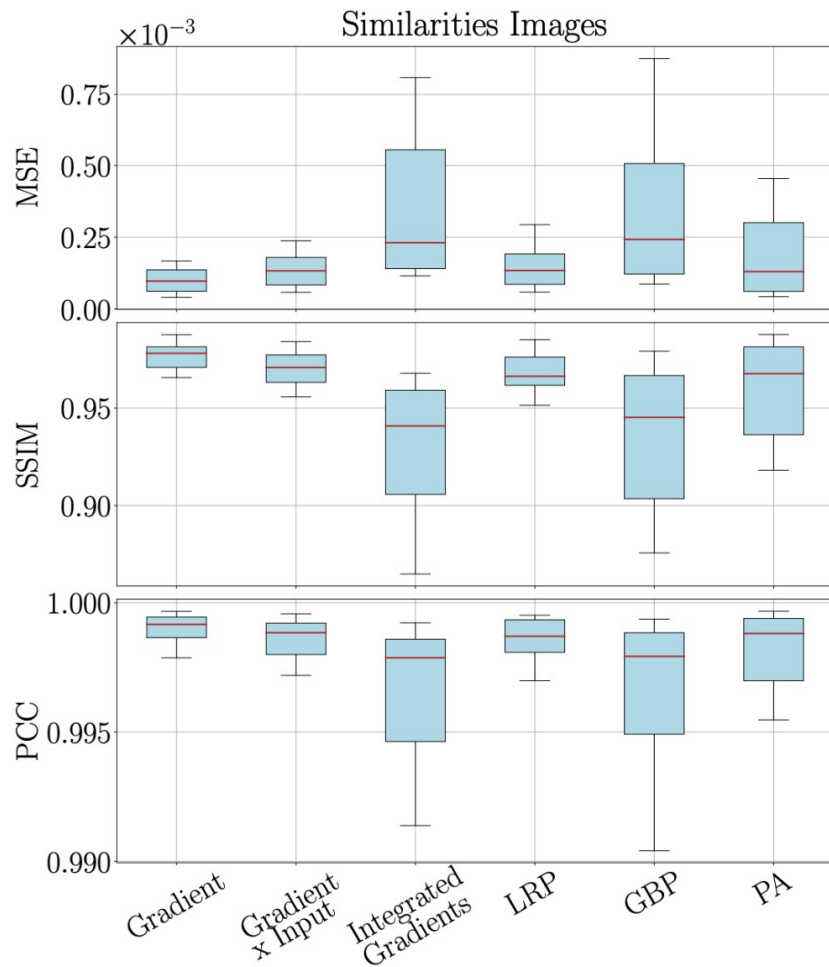
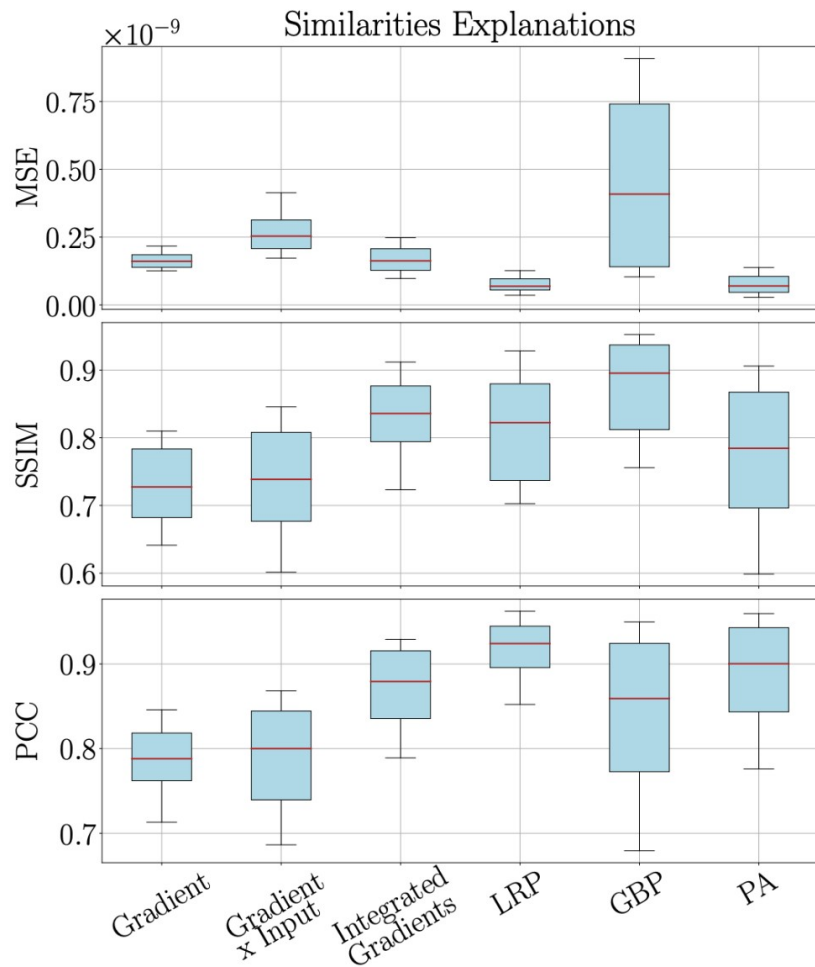
Experiments



Experimental Setup

- Apply algorithm to 100 randomly selected images for each explanation method
- Use VGG-16 network pre-trained on ImageNet
- For each run, we randomly select two images from the test set.
 - One of the two images is used to generate a target explanation map
 - The other image is perturbed by our algorithm with the goal of replicating the target using a few thousand iterations of gradient descent
- Comparable results obtained for ResNet-18, AlexNet, and Densenet-121 + CIFAR-10 dataset

	Original Map	Target Map	Manipulated Map	Perturbed Image	Perturbations	
Gradient						
Gradient x Input						
Layerwise Relevance Propagation						
Integrated Gradients						
Guided Backpropagation						
Pattern Attribution						



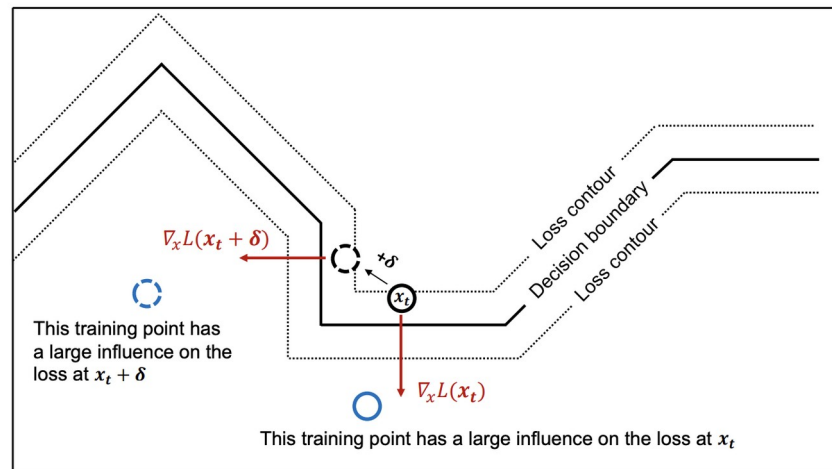
Theoretical Analysis

Intuition

Why are explanations vulnerable and unreliable?

[Ghorbani et al.]

Large curvature of the NN output manifold!



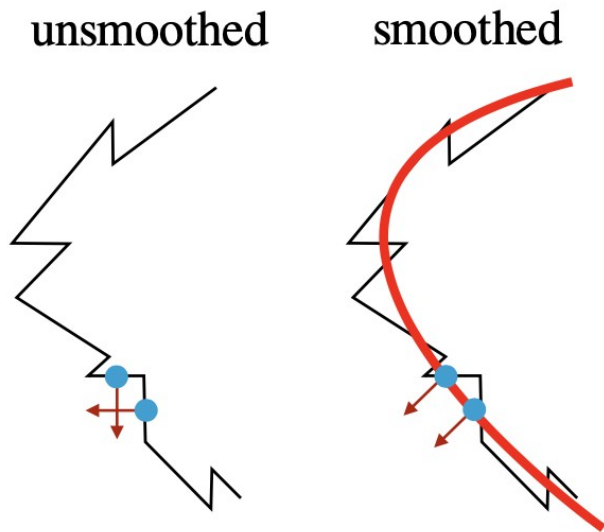
Theoretical Bound

Theorem 1 *Let $g : \mathbb{R}^d \rightarrow \mathbb{R}$ be a network with softplus_β non-linearities and $\mathcal{U}_\epsilon(p) = \{x \in \mathbb{R}^d; \|x - p\| < \epsilon\}$ an environment of a point $p \in S$ such that $\mathcal{U}_\epsilon(p) \cap S$ is fully connected. Let g have bounded derivatives $\|\nabla g(x)\| \leq c$ for all $x \in \mathcal{U}_\epsilon(p) \cap S$. It then follows for all $p_0 \in \mathcal{U}_\epsilon(p) \cap S$ that*

$$\|h(p) - h(p_0)\| \leq |\lambda_{\max}| d_g(p, p_0) \leq \beta C d_g(p, p_0), \quad (9)$$

where $\lambda_{\max}$ is the principle curvature with the largest absolute value for any point in $\mathcal{U}_\epsilon(p) \cap S$ and the constant $C > 0$ depends on the weights of the neural network.

Robustness via smoothing



$$\text{softplus}_{\beta}(x) = \frac{1}{\beta} \log(1 + e^{\beta x})$$

Smoothing: Connections to SmoothGrad

Theorem 2 For a one-layer neural network $g(x) = \text{relu}(w^T x)$ and its β -smoothed counterpart $g_\beta(x) = \text{softplus}_\beta(w^T x)$, it holds that

$$\mathbb{E}_{\epsilon \sim p_\beta} [\nabla g(x - \epsilon)] = \nabla g_{\frac{\beta}{\|w\|}}(x),$$

where $p_\beta(\epsilon) = \frac{\beta}{(e^{\beta\epsilon/2} + e^{-\beta\epsilon/2})^2} \cdot$

SmoothGrad

β -smoothing

$$\epsilon_i \approx \mathcal{N}(0, \sigma) \text{ with variance } \sigma = \log(2) \frac{\sqrt{2\pi}}{\beta}$$

Robustness experiments

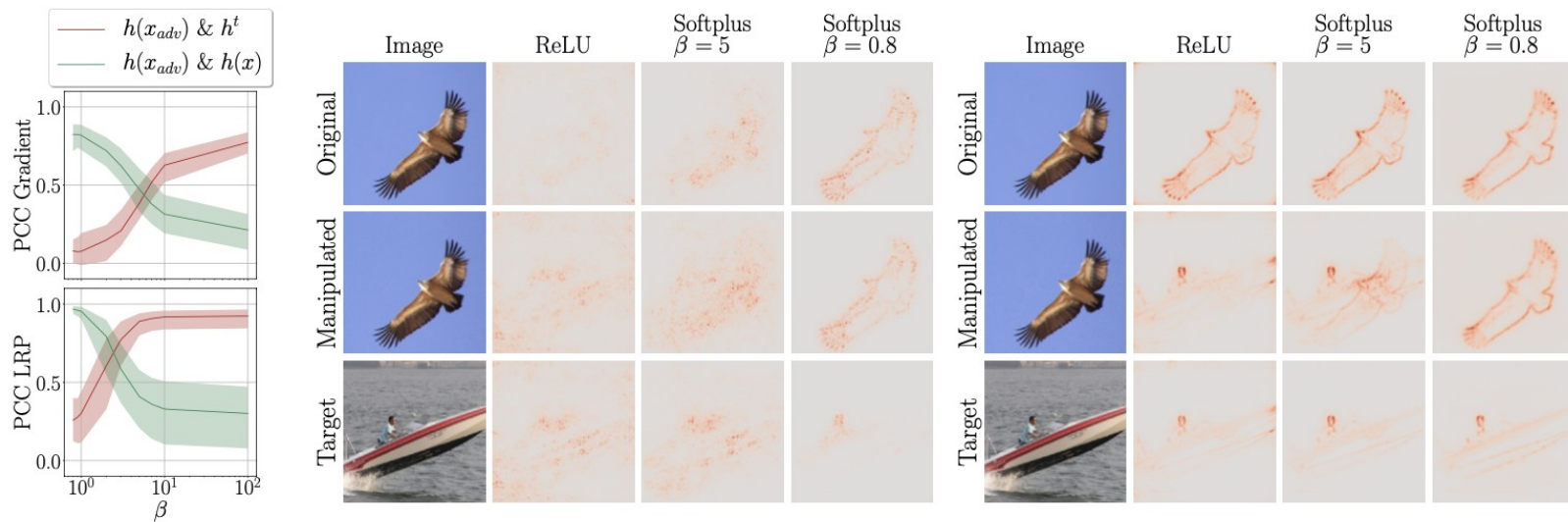


Figure 4. β -smoothing makes explanations more robust.

Robustness experiments

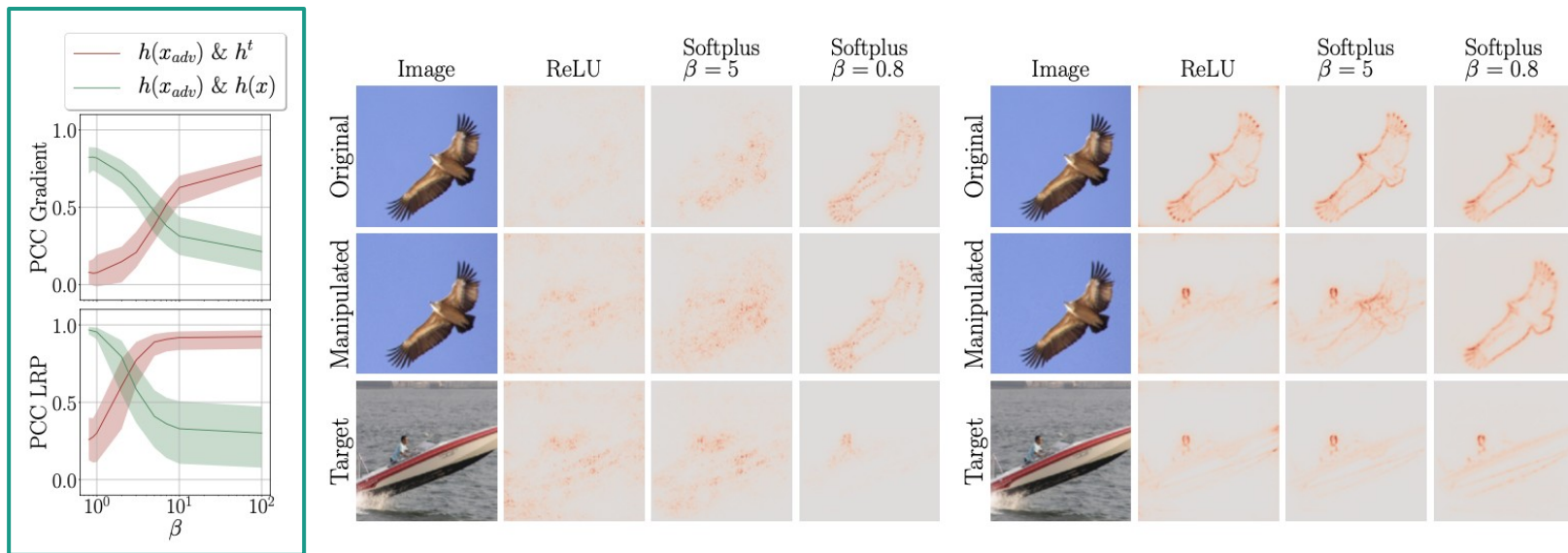


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Robustness experiments

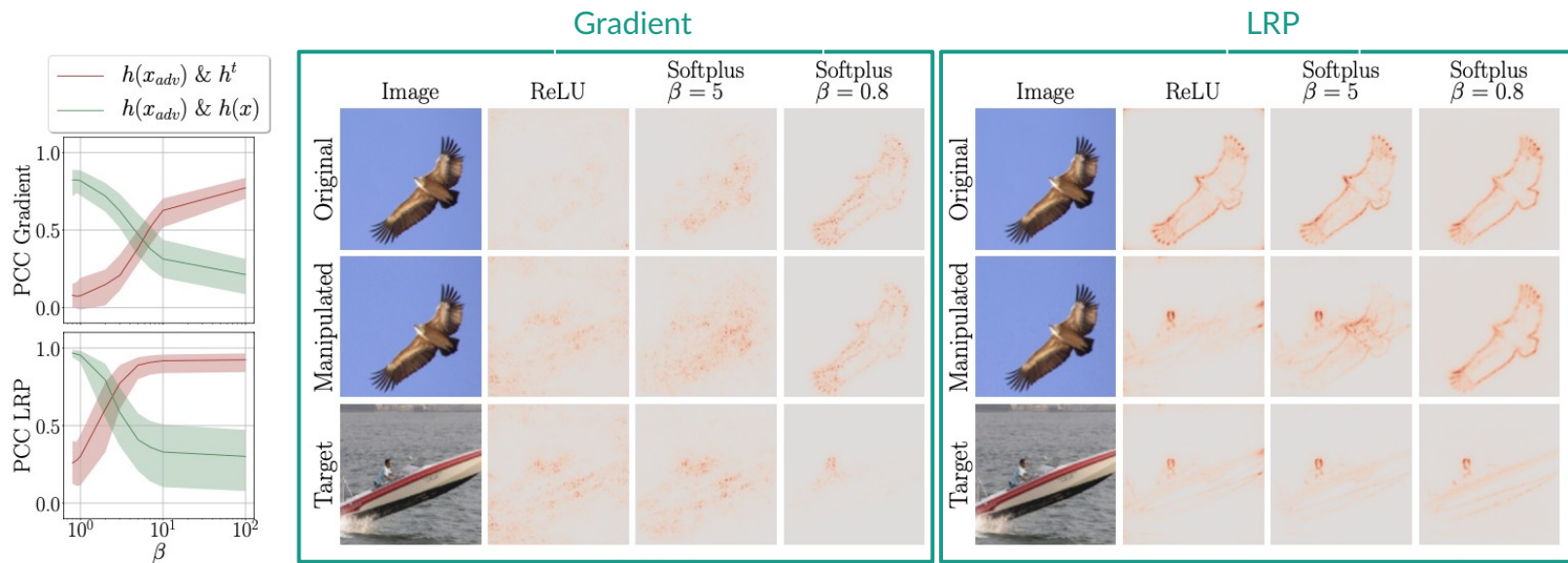


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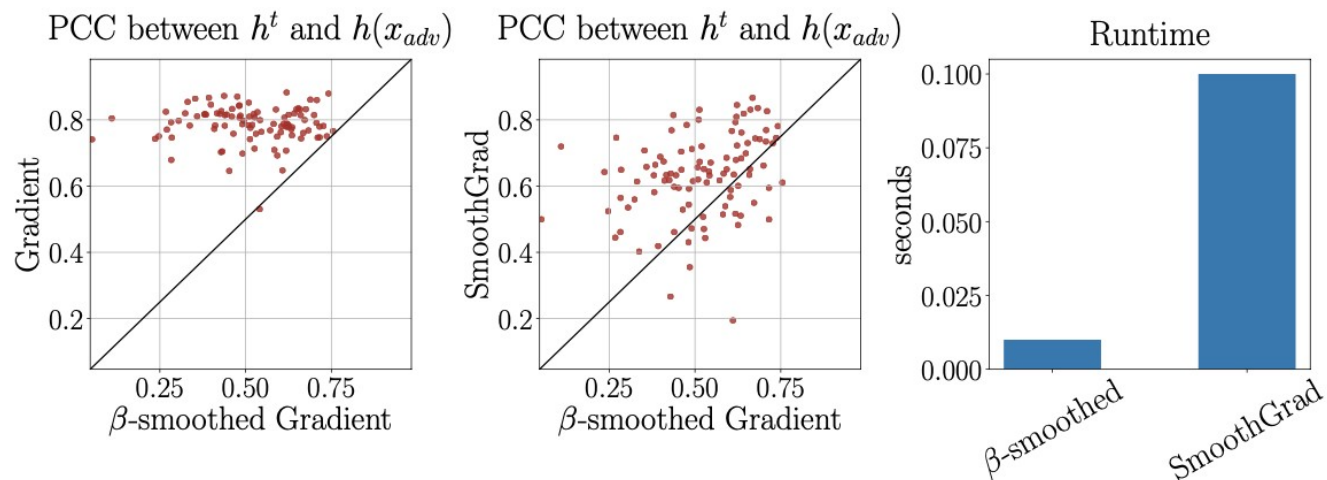


Figure 5. β -smoothing 1) makes explanations more robust
2) is comparable to SmoothGrad
3) has a faster runtime than SmoothGrad

Robustness experiments

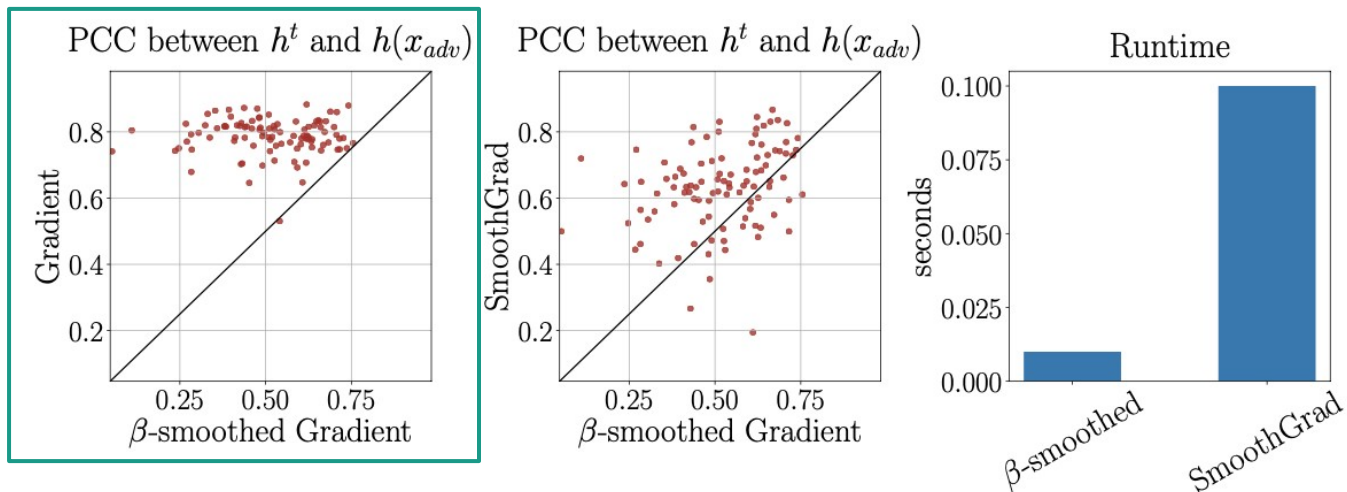


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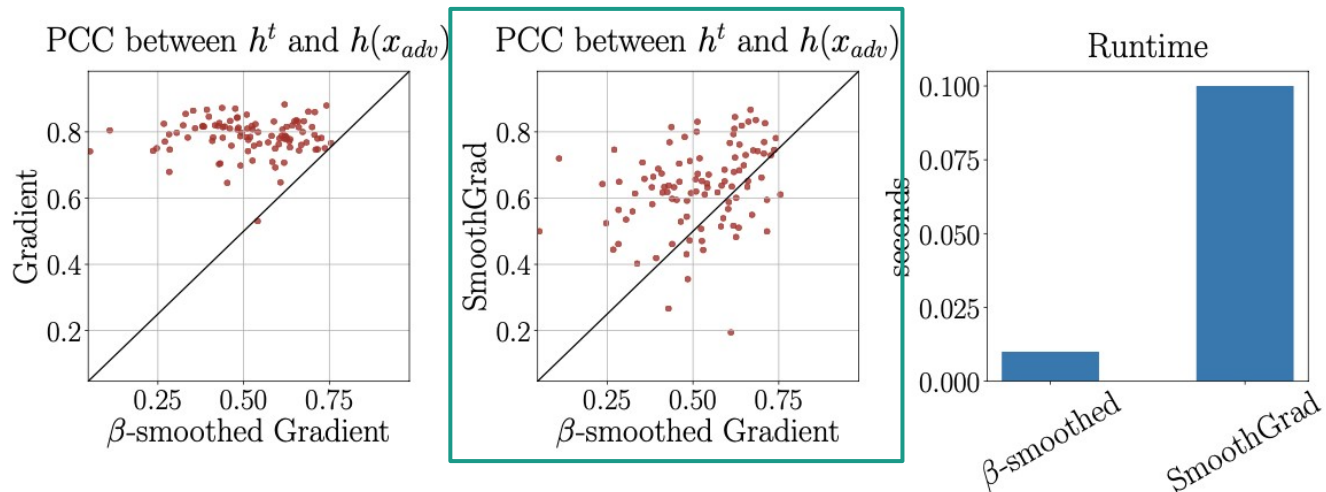


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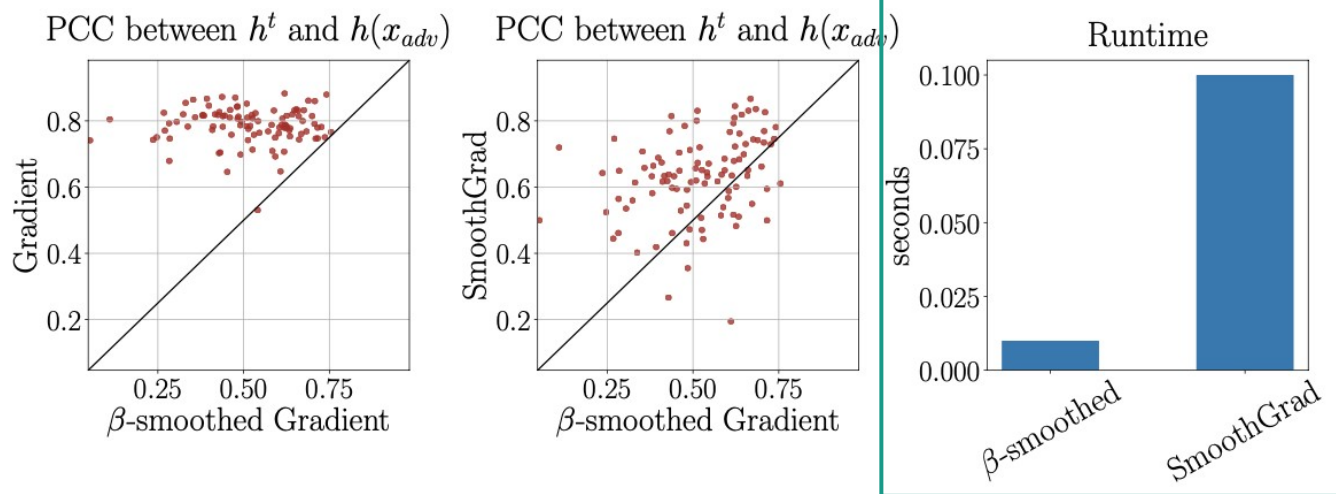


Figure 5. β -smoothing 1) makes explanations more robust
2) is comparable to SmoothGrad
3) has a faster runtime than SmoothGrad

Conclusion



Critique

Strengths

- Thorough investigation: problem → reason → solution
- Extensive validation: various explanation methods, models, and datasets

Limitations

- Analyses focused on relu/softplus activation function
- Evaluation of robustness based on Pearson correlation coefficient



Future directions & food for thought

Future directions

- Extend empirical analyses to other tasks and data modalities
- Generalize theoretical analyses to propagation-based methods
- Modify model training process to make NNs less vulnerable to explanation manipulation
 - Low-curvature models [Srinivas et al., NeurIPS 2022]

Food for thought

- Might there be other reasons for explanations being sensitive to manipulation?
- What are other ways to evaluate robustness of explanations?
- Is it a good idea to trade-off faithfulness for better robustness?